Risk Management Protocol

Context

Drones that aid people on big events like Aalst carnival. This will help our stakeholders, being the hospitals and the paramedic companies.

Risk assessment:

1. Malfunction
2. External factors like building, birds and other animals - collisions
3. Weather
4. Inaccurate report from the drone
5. Dataleaks→ privacy
6. Hacking -> taking over the drone
7. Theft

No	Risks	Category	Likelihood (1-5)	Impact (1-5)	Risk Score	Mitigation Methods
1.	Device malfunction (hardware/software)	technical	2	4	8	Maintenance, stand by mechanic
2.	External factors		1	5	5	Anticollision sensors
3.	Extreme weather		3	4	12	Waterproof, stabilizer
4.	Inaccurate report from the drone	Technical	1	4	4	Verification by medical personnel
5.	Dataleaks		1	2	2	Limit retention period, strong encryption

6.	Hacking		1	5	5	Monitoring, firewall, regular updates, Emergency failsafe procedures
7.	Theft		1	5	5	GPS, markings and audio warnings

Risk identification, assessment and preparation

Risk 1: software or hardware malfunction

Risk Identification

Risk: Software or hardware malfunction in a semi-autonomous drone

Description:
Failures in flight-control software, sensors, communication systems, or physical components (motors, batteries) can lead to loss of control, unsafe navigation, overuse of defect or crashes.

Causes:

- Software bugs or faulty updates
- Sensor inaccuracies or failures
- Battery, motor, or processor degradation
- Communication loss with the operator
- Environmental stress (weather, interference)

Consequences:

- Injury to people
- Property damage
- Loss of drone
- Legal and regulatory violations

Risk Assessment

Aspect	Level
Likelihood	2
Impact	4
Overall Risk	8

Justification:
Semi-autonomous systems are technically complex and operate in dynamic environments. Failures can result in severe safety and legal consequences

Risk treatment

Reduce

- Create a signal where the drone has crashed.
- Built-in system that detects when the drone is unstable and commands it to go back.
- Provided back-up drone
- Regular maintenance

Risk 2 Collision Risk – External Factors

Risk Identification

Risk: The semi-autonomous drone may collide with external obstacles such as buildings, infrastructure (e.g. power lines), birds, or animals due to unpredictable movement, limited sensor range, or environmental conditions like wind or low visibility.

Causes:

- Unpredictable movement of birds and animals entering the flight path.
- Dense or complex environments (urban buildings, trees, power lines).
- Sensor limitations (blind spots, limited range, poor performance in rain, fog, or low light).
- Environmental effects such as wind gusts causing drift toward obstacles.
- Insufficient pre-flight assessment of surroundings and temporary obstacles.
- Software limitations in obstacle recognition or path-planning algorithms.

- Human factors, including delayed manual intervention or overreliance on autonomy.

Consequences:

- Damage or total loss of the drone due to impact
- Injury to people or animals in the collision area
- Damage to buildings, vehicles, or infrastructure
- Mission failure or interruption (e.g. incomplete data collection or delivery)
- Legal and financial consequences, including liability claims or fines
- Loss of trust from stakeholders, regulators, or the public
- Operational downtime due to repairs, investigations, or grounding

Risk Assessment

Aspect	Level
Likelihood	1
Impact	5
Overall Risk	5

Risk treatment

Reduce

- Collision avoidance sensors & software
- Train the drone to identify the situation precisely.

Risk 3: extreme weather

Risk Identification

Description:

The semi-autonomous drone may be exposed to extreme weather conditions such as strong wind, heavy rain, fog, snow, or extreme temperatures, which can negatively affect flight stability, sensor performance, and overall system reliability.
Causes:

- Strong wind gusts and turbulence exceeding flight tolerance

- Rain, snow, or hail interfering with motors, electronics, or sensors
- Fog or low light reducing visual and sensor accuracy
- Extreme temperatures affecting battery performance and electronics
- Rapid weather changes not detected during pre-flight checks

Consequences:

- Loss of flight stability or navigation accuracy
- Collision or uncontrolled landing/crash
- Damage to drone components (motors, battery, sensors)
- Mission delay, abortion, or failure
- Increased safety risk to people and property
- Shortened drone lifespan due to environmental stress

Risk Assessment

Aspect	Level
Likelihood	3
Impact	4
Overall Risk	12

Risk treatment

Avoid

- No flying if the weather is not inside safe parameters drones can get broken and will take a lot of costs. It's up to the paramedics.

Share

- Insurance if services are critical for the event.

Reduce

- Stronger and more durable drones.

Risk 4: Inaccurate Reports or Views from the Drone

Risk Identification

Description:
The semi-autonomous drone may generate inaccurate, incomplete, or misleading reports or visual data due to sensor errors, environmental interference, data processing issues, or transmission problems, leading to incorrect interpretation or decision-making.

Causes

- Sensor calibration errors or degraded camera/sensor hardware
- Poor visibility conditions (glare, fog, shadows, low light) affecting image quality
- GPS inaccuracies or signal multipath effects in urban environments
- Data transmission loss, delay, or compression artifacts
- Software or AI misinterpretation of sensor data
- Insufficient data validation or redundancy in reporting systems

Consequences

- Incorrect situational awareness for operators or decision-makers
- Faulty or unsafe operational decisions based on incorrect data
- Mission inefficiency or failure due to unreliable outputs
- Loss of trust in drone-generated data and reports
- Potential legal or regulatory issues if incorrect data is relied upon
- Rework or repeated missions, increasing cost and risk

Risk Assessment

Aspect	Level
Likelihood	1
Impact	4
Overall Risk	4

Risk Treatment

Reduce

- A medical professional is on standby that verifies what the drone sees and documents.

Share

- Medical insurance

Risk 5: Data Leak

Risk Identification

Description:
The semi-autonomous drone may suffer a data leak, where sensitive data such as video footage, images, telemetry, or location information is accessed, intercepted, or disclosed to unauthorized parties during collection, transmission, storage, or processing.

Causes

- Unencrypted or weakly encrypted data transmission between drone and ground station
- Insecure cloud storage or backend systems
- Unauthorized access to operator devices or control stations
- Misconfigured access controls or permissions
- Insider misuse or human error (e.g. sharing data unintentionally)
- Software vulnerabilities in drone firmware or data-processing systems

Consequences

- Violation of privacy of individuals or organizations
- Non-compliance with data protection regulations (e.g. GDPR)
- Legal and financial penalties
- Loss of public and stakeholder trust
- Misuse of leaked data (surveillance, profiling, or malicious intent)
- Reputational damage to the operating organization

Risk Assessment

Aspect	Level
Likelihood	1
Impact	4
Overall Risk	4

Risk Treatment

Reduce

- All the data on the drone gets deleted after a short period of time.

Avoid

- No storage or processing of data that is not needed for services.

Risk 6: Hacking / Drone Interception

Risk Identification

Description

The semi-autonomous drone may be hacked or intercepted by unauthorized parties, allowing attackers to take control of the drone, manipulate its behavior, disrupt operations, or access onboard systems and data.

Causes

- Weak or outdated authentication mechanisms for command-and-control links
- Unencrypted or poorly secured communication channels (RF, Wi-Fi, LTE/5G)
- GPS spoofing or signal jamming attacks
- Software vulnerabilities in drone firmware or ground control software
- Use of default credentials or insecure configuration
- Lack of intrusion detection or monitoring systems

Consequences

- Loss of control over the drone (hijacking or forced landing)
- Collision, crash, or intentional misuse of the drone
- Threats to public safety and critical infrastructure

- Unauthorized surveillance or data manipulation
- Legal liability and regulatory violations
- Severe reputational damage to the operating organization

Risk Assessment

Aspect	Level
Likelihood	1
Impact	5
Overall Risk	5

Risk Treatment

Reduce

- Implement strong encryption and mutual authentication.
- Regularly update software.
- Monitor for intrusion in real time.
- Emergency failsafe procedures (return-to-home, auto-land)

Risk 7: Theft

Risk Identification

Description
The semi-autonomous drone may be stolen during storage, transport, deployment, or after landing, resulting in loss of equipment, data exposure, or potential misuse of the drone and its systems.

Causes

- Unsecured storage or transport of the drone and accessories
- Inadequate physical security at launch or landing sites
- Forced landing due to malfunction, low battery, or emergency situations
- Lack of tracking or recovery mechanisms
- Insider theft or unauthorized access by third parties

Consequences

- Financial loss due to stolen hardware and accessories
- Unauthorized access to onboard data or systems
- Reverse engineering or malicious reuse of the drone
- Operational disruption or mission failure
- Legal liability if stolen drone is misused
- Reputational damage to the operating organization

Risk Assessment

Aspect	Level
Likelihood	1
Impact	5
Overall Risk	5

Risk Treatment

Reduce

- Train the drone to recognize potential theft quickly.
- Loud audio warnings when the drone leaves the incident scene.
- Built-in GPS so the drone location can be tracked.